

Nifty 100 Financial Intelligence Platform

Analyst Guide

Sprint 6 reference for dashboard usage, API access, financial intelligence outputs, validation, and operational checks.

Current platform scope	Status
Nifty 100 companies	92 companies in the current database
Broad sectors	10 sectors in the supplied data
Dashboard	8 Streamlit screens
REST API	FastAPI under /api/v1
Cluster archetypes	5 KMeans clusters
Company tearsheets	92 two-page PDFs
Sector reports	10 one-page PDFs
Automated tests	221+ currently passing

Important data note: the source database currently contains 10 broad sectors, even though an earlier sprint specification mentioned 11. This guide follows the actual supplied dataset and does not invent an eleventh sector.

1. Platform Overview

The platform combines a normalized SQLite financial dataset, analytics modules, a Streamlit dashboard, FastAPI services, and generated analyst reports. The same database supports both dashboard exploration and API responses.

Logical architecture

Layer	Purpose	Primary location
Source data	Supplied Excel workbooks	data/raw/
ETL	Header handling, normalization, loading	src/etl/
Analytics	CAGR, ratios, cash-flow intelligence, clustering	src/analytics/
Database	SQLite analytical store	nifty100.db
Dashboard	Interactive analyst views	src/dashboard/
API	REST access to company and portfolio data	src/api/
Reports	Tearsheets, sectors, portfolio, charts	reports/
Tests	Unit, integration, performance	tests/

Core dashboard screens

Screen	Analyst use
Home	Portfolio KPIs, sector breakdown, top composite-score companies
Company Profile	Company-level KPIs, history, charts, pros/cons
Screener	Financial, growth, valuation and leverage filtering; CSV export
Peer Comparison	Peer-group metrics, benchmark comparison and radar view
Trend Analysis	Historical trend exploration across selected metrics
Sector Analysis	Sector medians, growth, ROE and distribution views
Capital Allocation	Cash-flow pattern and capital-allocation classification
Annual Reports	Annual-report years and external report links

2. Data Model and Coverage

The SQLite database is the operational source for dashboard and API queries. The companies table uses the ticker as its id; throughout the platform this value is exposed as company_id or ticker depending on the interface.

Key tables

Table	Key fields / role
companies	id, company_name, ROE, ROCE, website/profile links
sectors	company_id, broad_sector, sub_sector, index_weight_pct, market_cap_category
financial_ratios	year, profitability, leverage, cash-flow and growth ratios
profitandloss	sales, operating profit, net profit, EPS, dividend
balancesheet	capital, reserves, borrowings, assets and liabilities
cashflow	operating, investing, financing and net cash flow
market_cap	market cap, EV, P/E, P/B, EV/EBITDA, dividend yield
peer_groups	peer group membership and benchmark flag
peer_percentiles	peer metric values and percentile ranks
documents	annual report URLs by company and year
prosandcons	generated qualitative pros and cons

Known coverage characteristics

- Companies: 92.
- financial_ratios: 1,073 rows; latest available year is 2024 for the 92-company coverage used by the APIs.
- Market-cap history: 2019-2024 in the current database.
- Historical tables can contain duplicate year rows for some companies; API responses preserve the underlying data unless the endpoint explicitly selects the latest row.
- Some companies have limited historical depth and some annual report URLs are unavailable in source data.

3. Financial KPI Interpretation

Use KPIs as descriptive measures of historical financial performance. The platform is designed for comparative analysis, screening, and evidence review rather than as a standalone investment recommendation engine.

Core formulas

Metric	Formula / interpretation
Net Profit Margin	$\text{Net profit} / \text{sales} \times 100$
ROE	Return on equity percentage supplied/calculated in the ratio layer
ROA	Return on assets percentage
Debt to Equity	$\text{Debt} / \text{equity}$
Asset Turnover	$\text{Sales} / \text{assets}$
Revenue CAGR 5Y	$((\text{latest sales} / \text{sales five years earlier})^{(1/5)} - 1) \times 100$, when valid
PAT CAGR 5Y	Same method using net profit
FCF Margin	$\text{Free cash flow} / \text{revenue} \times 100$
CFO / PAT	$\text{Operating cash flow} / \text{net profit}$
CapEx Intensity	$\text{Absolute investing cash flow} / \text{revenue} \times 100$
FCF Conversion	$\text{Free cash flow} / \text{EBITDA}$
CFO / Total Debt	$\text{Operating cash flow} / \text{total debt}$
Investing / CFO	$\text{Investing cash flow} / \text{operating cash flow}$
Financing / CFO	$\text{Financing cash flow} / \text{operating cash flow}$

Interpretation cautions

- CAGR requires a positive starting value for the platform calculation; otherwise the result may be unavailable.
- Financial-sector cash-flow measures can behave differently from industrial companies, so cross-sector comparisons should be interpreted carefully.
- Extreme ROE or CAGR values can arise from small bases or data characteristics. The clustering workflow reports them instead of silently deleting them.
- Median statistics are often more robust than means when the portfolio contains extreme values.

4. Screener Workflow

The Streamlit Screener combines latest ratio data, market valuation data, sector mapping, and 5-year CAGR information. Results can be exported as CSV from the dashboard.

Available dashboard filters

Filter	Meaning
ROE minimum	Minimum latest return on equity percentage
D/E maximum	Maximum latest debt-to-equity ratio
FCF minimum	Minimum latest free cash flow in crore
Revenue CAGR 5Y minimum	Minimum five-year revenue CAGR
PAT CAGR 5Y minimum	Minimum five-year profit CAGR
OPM minimum	Minimum operating profit margin
P/E maximum	Maximum latest P/E ratio
P/B maximum	Maximum latest P/B ratio
Dividend Yield minimum	Minimum latest dividend yield
Interest Coverage minimum	Minimum latest interest coverage

Preset concepts

- Quality: emphasizes ROE, leverage control and operating quality.
- Value: emphasizes valuation ceilings with baseline quality checks.
- Growth: emphasizes revenue and PAT growth thresholds.
- Dividend: emphasizes dividend yield with baseline financial filters.
- Debt-Free: emphasizes very low debt-to-equity.
- Turnaround: emphasizes positive growth and cash-flow screens.

Analyst workflow: start broad, record the number of matches, tighten one filter at a time, and export the resulting CSV so the exact screen is reproducible.

5. REST API Reference

The FastAPI service is available under the `/api/v1` prefix. Interactive documentation is exposed at `/docs` and the OpenAPI document is available at `/openapi.json`.

Endpoint map

Area	Endpoint	Use
Companies	GET <code>/api/v1/companies</code>	List all 92 companies
Companies	GET <code>/api/v1/companies/{ticker}</code>	Company profile + latest KPIs
Companies	GET <code>/api/v1/companies/{ticker}/pl</code>	P&L history
Companies	GET <code>/api/v1/companies/{ticker}/bs</code>	Balance-sheet history
Companies	GET <code>/api/v1/companies/{ticker}/cashflow</code>	Cash-flow history
Companies	GET <code>/api/v1/companies/{ticker}/ratios</code>	Ratio history
Companies	GET <code>/api/v1/companies/{ticker}/tearsheet</code>	PDF tearsheet response
Screener	GET <code>/api/v1/screener</code>	Filtered company set
Sectors	GET <code>/api/v1/sectors</code>	Sector counts + medians
Sectors	GET <code>/api/v1/sectors/{sector}/companies</code>	Companies in sector
Peers	GET <code>/api/v1/peers/{group_name}</code>	Peer membership + percentile rows
Peers	GET <code>/api/v1/companies/{ticker}/peers/compare</code>	8-metric peer comparison
Valuation	GET <code>/api/v1/valuation</code>	Latest valuation snapshot
Valuation	GET <code>/api/v1/market-cap/{ticker}</code>	Market-cap history
Portfolio	GET <code>/api/v1/portfolio/stats</code>	Portfolio KPI statistics
Documents	GET <code>/api/v1/companies/{ticker}/documents</code>	Annual reports + URL status
Health	GET <code>/api/v1/health</code>	Service and DB health

6. API Usage and Response Notes

Start the API from the project root with the project virtual environment activated.

Run command

```
python -m uvicorn src.api.main:app --port 8000
```

Useful browser checks

- <http://127.0.0.1:8000/docs> - interactive Swagger UI.
- <http://127.0.0.1:8000/openapi.json> - OpenAPI schema.
- <http://127.0.0.1:8000/api/v1/health> - JSON health response.

Representative health response fields

Field	Meaning
status	Service state; current verified value is ok
db_row_counts	Row counts for major database tables
uptime_seconds	Approximate API process uptime
version	API version string; current service is 1.0.0

Example screener request

```
GET /api/v1/screener?min_roe=10&max;_de=2
```

The current implementation returns an object with count and data. Individual rows expose ticker, company_name, broad_sector, market_cap_category, ROE, D/E, FCF, CAGR metrics and P/E where available.

Example company profile note: `/api/v1/companies/TCS` returns company, sector and latest_kpis objects rather than a flat ticker field at the top level.

7. Peer, Valuation and Portfolio Analysis

Peer comparison supports relative analysis inside defined peer groups. The current comparison response contains eight latest-year metrics and exposes the selected company, the peer average, and the benchmark value.

Peer metric	Purpose
ROE	Profitability relative to peer set
D/E	Relative leverage
Revenue Growth	Latest growth comparison
Net Margin	Profit conversion comparison
OPM	Operating efficiency comparison
Interest Coverage	Interest-servicing capacity
FCF Margin	Cash generation relative to revenue
CapEx Intensity	Investment intensity relative to revenue

Valuation endpoint fields

- Latest market capitalization, P/E, P/B, EV/EBITDA and dividend yield.
- Latest free cash flow and an FCF-yield calculation where the required values exist.
- Simple P/E flag categories are descriptive thresholds and should be interpreted alongside sector context and business model.

Portfolio stats

The portfolio endpoint reports P10, P25, P50, P75, P90, Mean, Std and sample_count for eight latest-year portfolio KPIs. Because some distributions are highly skewed, the median and percentile spread can be more informative than the mean alone.

8. Cash Flow Intelligence

The cash-flow module classifies each company using historical CFO quality, CapEx intensity, FCF behavior, distress conditions, deleveraging behavior and cash-flow sign patterns.

CFO quality

Average 5Y CFO/PAT	Label
> 1.0	High Quality
0.5 to 1.0	Moderate
< 0.5	Accrual Risk
Unavailable	N/A

CapEx intensity

CapEx intensity	Label
< 3%	Asset Light
3% to 8%	Moderate
> 8%	Capital Intensive

Capital allocation patterns

CFO / CFI / CFF signs	Pattern
+ / - / -	Self-Funded Growth
+ / - / +	Growth + External Funding
+ / + / -	Shareholder Returns
+ / + / +	Cash Accumulator
- / - / +	Distress / Funding Need
- / + / +	Asset Monetisation
- / - / -	Cash Burn
- / + / -	Restructuring

9. Clustering and Archetypes

The clustering workflow applies KMeans with five clusters and random_state=42 after median imputation and standardization. Five features are used:

- Return on equity percentage
- Debt to equity
- Revenue CAGR 5Y
- FCF CAGR 5Y
- Operating profit margin percentage

Outputs

Output	Purpose
output/cluster_labels.csv	Company-to-cluster assignments and centroid distance
output/cluster_profile.csv	Cluster-level statistics
output/outlier_report.csv	Detected outlier rows
output/portfolio_stats.csv	Portfolio statistics for clustering features
reports/elbow_plot.png	K selection diagnostic for k=2..10
reports/correlation_heatmap.png	Pearson relationship view across requested KPIs

Current five archetype labels

Cluster id	Label	Current company count
0	Turnaround / Cyclical	60
1	Defensive Quality	15
2	Value / Balanced	2
3	High-Quality Compounders	14
4	Emerging Growth	1

Small clusters should not be interpreted as statistically representative on their own; they are labels over the current 92-company dataset.

10. Reports and Validation Outputs

The reporting layer produces company tearsheets, sector reports, radar charts, and a 92-page portfolio summary. Day 44 documentation should make the output locations easy to discover and reproduce.

Output	Current state / purpose
reports/tearsheets/	92 company PDFs; each generated tearsheet is two pages in the current Sprint 5 output
reports/sector/	10 one-page sector PDFs matching the supplied broad-sector data
reports/portfolio/portfolio_summary.pdf	92 pages, one company per page
reports/radar_charts/	92 company peer radar images
output/pros_cons_generated.csv	466 generated pros/cons rows; all 92 companies have at least one pro and one con
output/cashflow_intelligence.xlsx	92-company cash-flow intelligence output
output/capital_allocation.csv	Historical capital-allocation classifications
sprint5_qa.md	Sprint 5 QA record and known data/spec mismatch

Validation principles

- Check file count and expected coverage before reviewing individual content.
- Check PDF page counts and file sizes to catch empty or incomplete outputs.
- Use visual rendering for layout-sensitive PDFs; text extraction alone cannot detect overlap or clipping.
- Keep data-quality exceptions documented rather than silently removing unusual rows.

11. Operational Runbook

The following sequence is a practical analyst/developer smoke test after a code or data change.

Step 1 - run automated tests

```
python -m pytest -q
```

Day 42 baseline: 221 tests passed. Day 43 added four targeted checks: three performance tests and one dashboard/API integration test. Rerun the full suite after Day 43 to refresh the exact total.

Step 2 - start API

```
python -m uvicorn src.api.main:app --port 8000
```

Step 3 - start dashboard in a second terminal

```
python -m streamlit run src\dashboard\app.py
```

Step 4 - smoke test

- Open /docs and confirm API routes appear.
- Check /api/v1/health returns HTTP 200.
- Open the Streamlit dashboard at <http://localhost:8501>.
- Open the Screener page and run a small filter.
- Open one company profile and one annual-report view.

Performance baseline from Sprint 6 Day 43

Check	Current verified result
10 concurrent screener requests	Under 10 seconds
Five company profile requests	Each under 3 seconds
FastAPI + Streamlit together	Verified; FastAPI health returned HTTP 200 while Streamlit was running
SQLite indexes	Indexes added for common company/year, sector, market-cap, document and peer queries

12. Troubleshooting and Known Limitations

Common operational issues

- Use `python -m streamlit` rather than plain `streamlit` when the shell does not recognize the command.
- Use `python -m pytest` for the project virtual environment so the correct interpreter and plugins are used.
- PowerShell can reinterpret nested quotes in `python -c` commands. Prefer a PowerShell here-string piped to `python` for multi-line SQLite checks.
- Keep API and Streamlit in separate terminals when testing simultaneous operation.

Known data-quality limitations

- Some historical tables contain duplicate company/year records. They were preserved and tested rather than silently deleted.
- Some source companies have fewer than 10 years of history.
- Annual-report data includes missing URLs for some older years.
- The source data currently maps to 10 broad sectors, not 11.
- Some cluster statistics contain extreme values because the supplied data includes very large ratios or growth values.
- The documents API marks URL structure validity; it does not by itself prove that a remote website or PDF is reachable.

Release-readiness checklist

Area	Check
Tests	Full pytest suite passes with zero failures
API	Health and core endpoints return expected status codes
Dashboard	All eight screens load without unhandled exceptions
Reports	Expected files exist and PDFs render without clipping
Data quality	Validation failures are reviewed and documented
Documentation	README, analyst guide and QA notes match current implementation

13. Analyst Glossary

Term	Meaning
CAGR	Compound annual growth rate over the specified period
CFO	Cash flow from operating activities
CFI	Cash flow from investing activities
CFF	Cash flow from financing activities
FCF	Free cash flow
PAT	Profit after tax / net profit
ROE	Return on equity
ROCE	Return on capital employed
OPM	Operating profit margin
P/E	Price-to-earnings ratio
P/B	Price-to-book ratio
EV/EBITDA	Enterprise value to EBITDA
CapEx	Capital expenditure
Peer group	Defined set of comparable companies
Percentile	Relative position of a value within a distribution
KMeans	Centroid-based clustering algorithm used for five archetypes
Tearsheet	Compact company-level PDF report
Validation failure	A row recorded by the DQ validator when a rule is violated

End of guide

Use this guide together with the project README, sprint QA notes, OpenAPI documentation, and the generated output files. When implementation details change, update the guide and README in the same commit so analyst instructions remain synchronized with the codebase.